Understanding the FitzHugh–Nagumo Model

Jose Luna and Peter Marandos

July 2024

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1 Introduction

In 1952, Alan Hodgkin and Andrew Huxley produced a model to describe the action potential of a single nerve cell (Hodgkin and Huxley, 1952). Although their model accurately captures the oscillatory behavior of neurons, it is a four-dimensional nonlinear system, and is thus very difficult to analyze. A simplification was made in 1961, when Richard FitzHugh proposed a two-dimensional variant based on the Van der Pol oscillator and showed that it captured the salient features of the Hodgkin-Huxley model: namely, the presence of relaxation oscillations and widely separated timescales and a rich set of dynamical behaviors and bifurcations from its dependence on parameters (FitzHugh 1961).

A year later, Jin-Ichi Nagumo et al. designed an electrical circuit and demonstrated that its dynamics were equivalent to FitzHugh's model (Nagumo et al. 1962); thus, this model is now known as the FitzHugh-Nagumo (or FHN) Model.

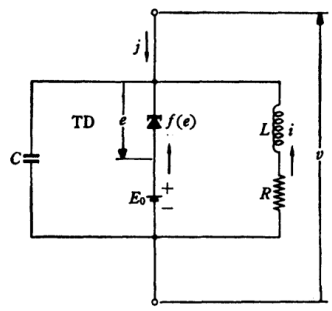

Figure 1: Equivalent circuit diagram from Nagumo's original paper

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Beyond neuroscience, these equations have found applications in modeling diverse phenomena such as cardiac cells (Keener & Sneyd 2010), turbulent chemical reactions (Kuramoto et al. 1998) and epilepsy (Moayeri et al. 2020).

In this project, we will analyze the dynamics of this model, including:

- Classification and stability of fixed points
- Bifurcation analysis for varying a, b and I
- Dynamical regimes for the system
- Physical and biological interpretations

2 The FitzHugh-Nagumo Model

The FHN Model is comprised of two nonlinear first-order differential equations (Sherwood 2014):

$$\dot{v} = v - \frac{v^3}{3} - w + I$$
$$\dot{w} = \epsilon(v + a - bw)$$

where:

- v represents the membrane potential of the neuron
- w is a "recovery variable" representing the combined effects of Na^+ channel reactivation and K^+ channel deactivation
- I is the current that is applied to the neuron

- a and b are positive constants controlling the dynamics of the system
- $\epsilon << 1$ is a positive constant that induces a separation of timescales, creating a "fast" variable (v) and a "slow" variable (w)

To get a sense for the kind of behavior that is possible, we can plot v and w versus t for parameter values of $a = 0.7$, $b = 0.8$, $\epsilon = 0.08$ and $I = 0.5$ (representing constant applied current). $v(t)$ is shown in blue, and $w(t)$ is shown in red:

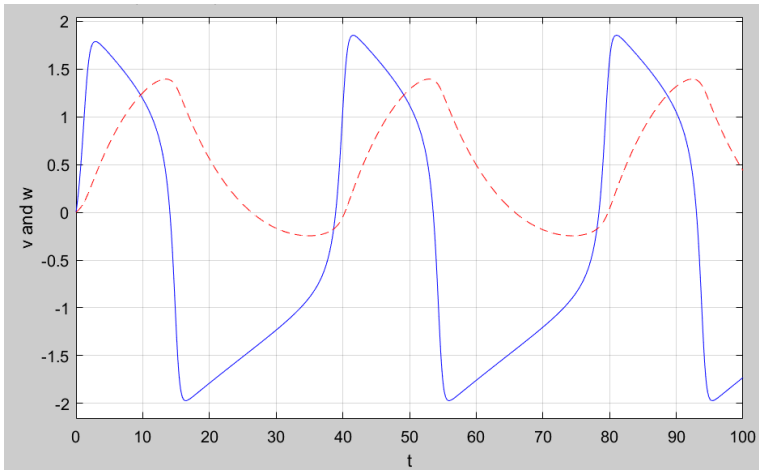

Figure 2: Initial Condition: $(0, 0)$

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As we can see, under constant applied current and certain parameter values, neurons exhibit oscillations in membrane potential, which represents a constant-rate firing. However, as we will see, the dynamics are heavily dependent upon parameter values. To begin our analysis, we will find and classify the fixed points of the system.

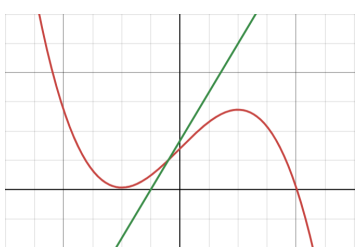
3 Fixed Points

To find the fixed points, we consider first the two nullclines $\dot{v} = 0$ and $\dot{w} = 0$, which give us curves:

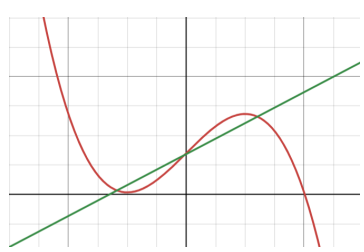
$$w = v - \frac{v^3}{3} + I \quad (v\text{-}nullcline)$$

$$w = \frac{v + a}{b} \quad (w\text{-}nullcline)$$

The fixed points are located at the intersection points of these nullclines in (v, w)-space. As we can see from the figure below, the nullclines will intersect at either one or three points, depending on the parameters:



(a) $I = 0.7$, $a = 0.5$, $b = 0.6$



(b) $I = 0.7$, $a = 1.3$, $b = 1.9$

Setting the nullcline equations equal to each other produces a cubic equation in v:

$$v - \frac{v^3}{3} + I = \frac{v+a}{b} \implies \frac{b}{3}v^3 - (b-1)v + a - bI = 0$$

It is difficult to find explicit roots; however, we can instead turn our attention to the *number of roots*. This cubic equation will only have 3 real roots when it has two distinct critical points. Setting its derivative (with respect to v) equal to 0:

$$\begin{aligned} \frac{d}{dv}\left(\frac{b}{3}v^3 - (b-1)v + a - bI\right) &= 0 \\ bv^2 - b + 1 &= 0 \\ v &= \pm\sqrt{\frac{b-1}{b}} \end{aligned}$$

For this to have two real solutions, we need $b > 1$. Thus, we are guaranteed that when $b < 1$, there will be only one fixed point. Note, however, that $b > 1$ is a necessary but *not sufficient* condition for the cubic to have 3 real roots. To determine the conditions under which there are 3 fixed points, we will need to conduct numerical experiments.

4 $b < 1$: One fixed point

Focusing our attention first on the case where $b < 1$ (where we are guaranteed to have one fixed point), let's consider its stability. Let (v_f, w_f) denote the fixed point. We can examine its stability by calculating the Jacobian of the system. Letting $f(v,w) = \dot{v}$ and $g(v,w) = \dot{w}$, the partial derivatives are:

$$\begin{aligned} \frac{\partial f}{\partial v} &= 1 - v^2 & \frac{\partial f}{\partial w} &= -1 \\ \frac{\partial g}{\partial v} &= \epsilon & \frac{\partial g}{\partial w} &= -b\epsilon \end{aligned}$$

So the Jacobian at (v_f, w_f) is:

$$J_{v_f, w_f} = \begin{bmatrix} 1 - v_f^2 & -1 \\ \epsilon & -b\epsilon \end{bmatrix}$$

This matrix has trace $\tau = 1 - v_f^2 - b\epsilon$ and determinant $\Delta = -b\epsilon(1 - v_f^2) + \epsilon$. For (v_f, w_f) to be stable, we need the trace to be negative, and the determinant to be positive:

$$\begin{aligned} \tau &< 0 & \Delta &> 0 \\ 1 - v_f^2 - b\epsilon &< 0 & -b\epsilon(1 - v_f^2) + \epsilon &> 0 \\ v_f^2 &> 1 - b\epsilon & bv_f^2 - b + 1 &> 0 \end{aligned}$$

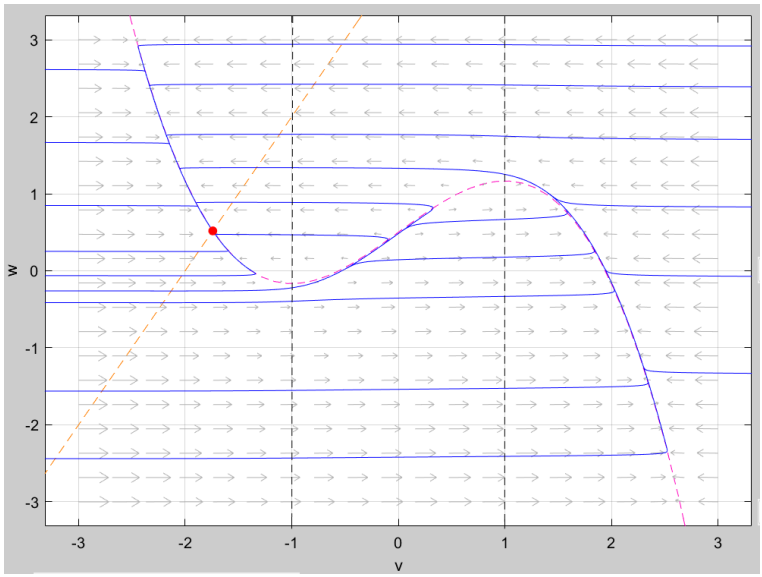
The determinant equation $bv_f^2 - b + 1 > 0$ will always be true since we are assuming $0 < b < 1$. So the condition for stability is:

$$v_f^2 > 1 - b\epsilon$$

In other, words, if $v_f \in (-\sqrt{1-b\epsilon}, \sqrt{1-b\epsilon})$, then the fixed point (v_f, w_f) will be unstable. We can think of v_f landing in one of three regions: $(-\infty, -\sqrt{1-b\epsilon})$, $(-\sqrt{1-b\epsilon}, \sqrt{1-b\epsilon})$ or $(\sqrt{1-b\epsilon}, \infty)$. FitzHugh in his original paper called these regions the *branches* of the $v-$nullcline. Whatever branch the fixed point is found in depends on the values of parameters a and I.

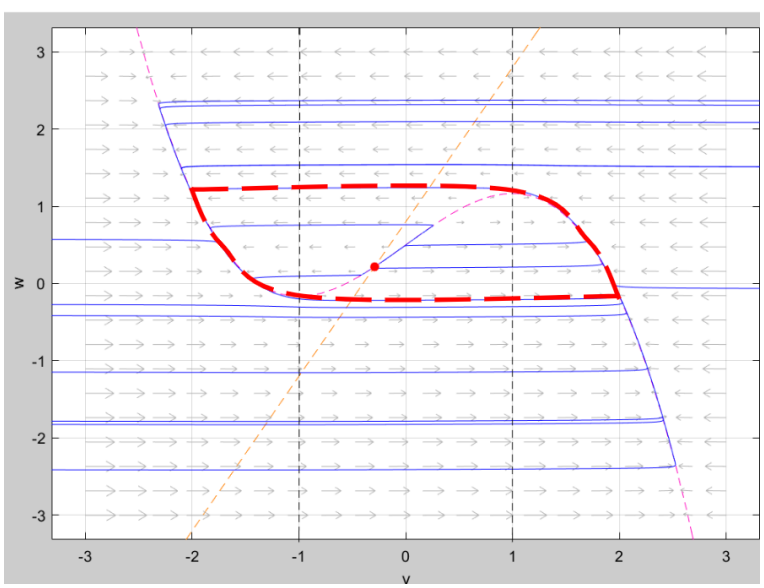
Let's graphically examine the behavior of the fixed point in each of these cases:

- $v_f < -\sqrt{1-b\epsilon}$ (**left branch**): Below is a plot of the system with parameters $a = 2$, $b = 0.5$, $I = 0.5$ and $\epsilon = 0.01$ for which the fixed point lands in the left branch. The fixed point has been highlighted in red, the nullclines are shown as colored dotted lines, and the branches have been demarcated by black dotted lines.

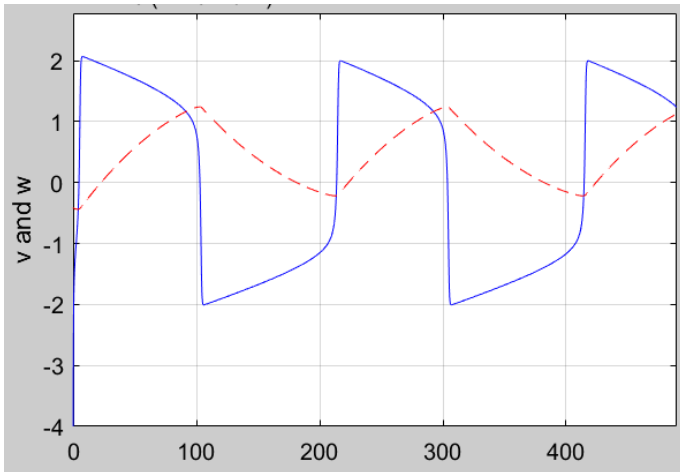


In this case, we find that trajectories very quickly approach the cubic nullcline, then travel along the nullcline until reaching the stable fixed point.

- $-\sqrt{1-b\epsilon} < v_f < \sqrt{1-b\epsilon}$ (**middle branch**): Below is the plot for parameters $a = 0.4$, $b = 0.5$, $I = 0.5$ and $\epsilon = 0.01$ for which the fixed point lands in the middle branch.

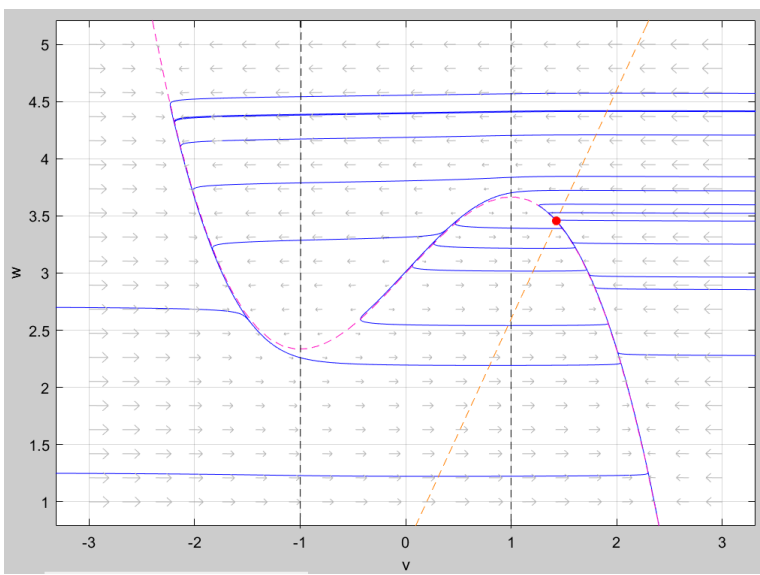


Here we see that trajectories approach the cubic nullcline extremely quickly in the horizontal direction, and slowly travel along the nullcline. However, in this case they settle into a stable limit cycle (highlighted in red).In fact, this is an example of a *relaxation oscillation*. Note that the fixed point has now become unstable, as expected. To better visualize the limit cycle, we also plot v and w vs. t for a certain trajectory:



Here the periodic behavior of v and w is apparent. The limit cycle appears to have a period of ≈ 200

- $v_f > \sqrt{1 - b\epsilon}$ (**right branch**): Below is a plot of the system with parameters $a = 0.3$, $b = 0.5$, $I = 3$ and $\epsilon = 0.01$ for which the fixed point lands in the right branch.



As expected, the fixed point has become stable again, and the limit cycle has disappeared.

4.1 Bifurcation and stability diagram

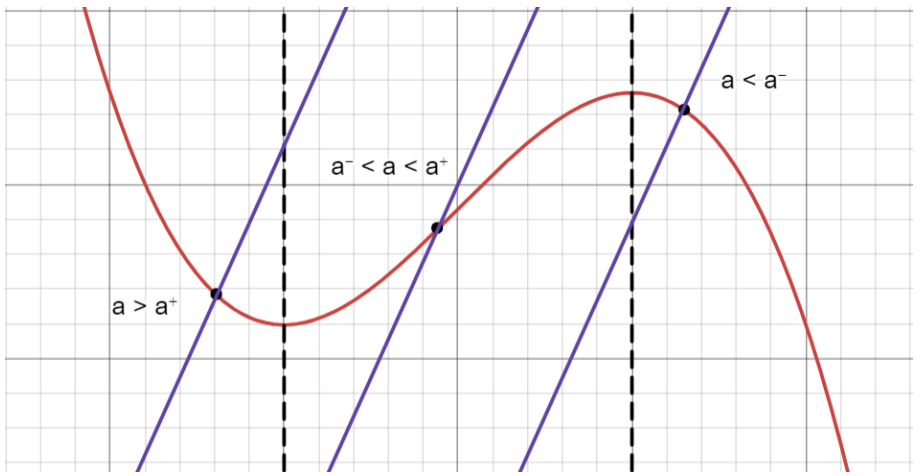
As we've seen, the branch that the fixed point lands on will be determined by the parameters a and I. If we take a closer look at the nullclines:

$$w = v - \frac{v^3}{3} + I$$
$$w = \frac{v + a}{b}$$

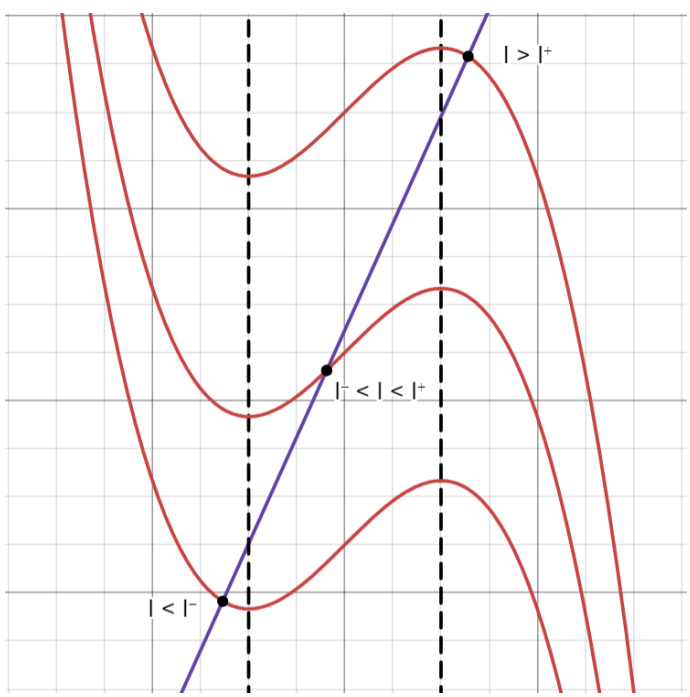
By inspection, we see that varying I will shift the cubic nullcline vertically, and varying a will shift the linear nullcline vertically.

Let's consider keeping I constant and starting with a low value of a such that the nullclines will intersect on the right branch. Then increasing a continuously will eventually cause the intersection point to cross over into the middle branch, making the fixed point unstable and creating a limit cycle. This is indicative of a *supercritical Hopf bifurcation*. As we continue increasing a, the intersection will cross into the left branch, causing another supercritical Hopf bifurcation (this time in reverse), which will render the fixed point stable again and destroy the limit cycle.

Thus, there will be two critical bifurcation values, a^- and a^+, such that the limit cycle exists when $a^- < a < a^+$. The argument is shown graphically below, where each linear nullcline arises from a different value of a (the branches are demarcated by dashed black lines):



By a similar argument, we can show that keeping a constant and starting with a low value of I will have an intersection point on the *left branch*, and increasing I continously will cause two supercritical Hopf bifurcations at I^- and I^+, as shown below:



How can we find these bifurcation values? An approximation will be handy. As we previously calculated, the bifurcations will occur when the $v-$coordinate of the fixed point, v_f, crosses into the range $(-\sqrt{1-b\epsilon}, \sqrt{1-b\epsilon})$. When $\epsilon << 1$ (as is assumed by the model), this range will be almost identical to the range $(-1, 1)$. Therefore, we can plug in the values $v_f = -1$ and $v_f = 1$ to our nullcline equations and set them equal to each other to obtain a relationship between a and I:

$$v_f - \frac{v_f^3}{3} + I = \frac{v_f + a}{b}$$

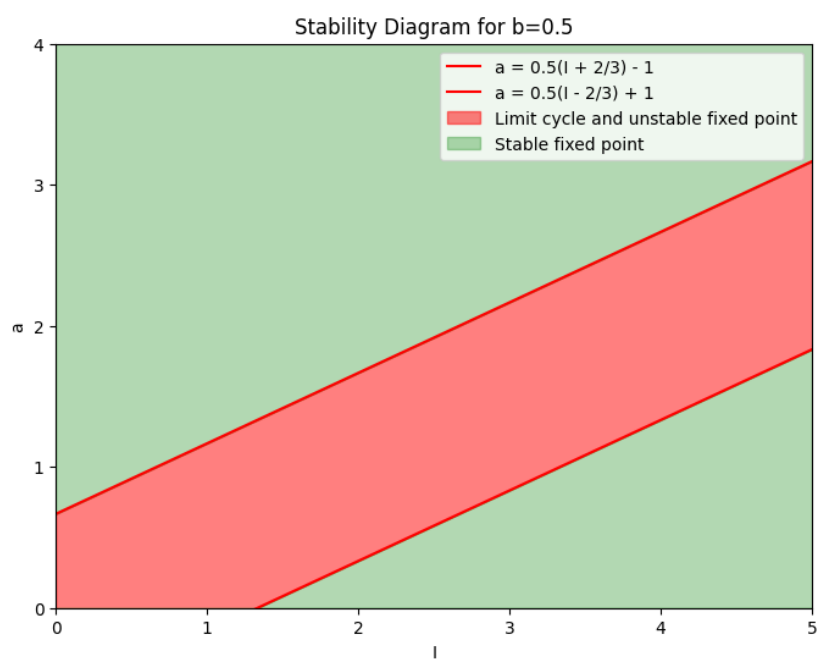
Setting $v_f = 1$ will determine the value for a^- in terms of I and b, or alternatively, the value for I^+ in terms of a and b:

$$\begin{aligned} 1 - \frac{1}{3} + I &= \frac{1+a}{b} \\ \implies a^- &= b(I + \frac{2}{3}) - 1 \\ \implies I^+ &= \frac{1+a}{b} - \frac{2}{3} \end{aligned}$$

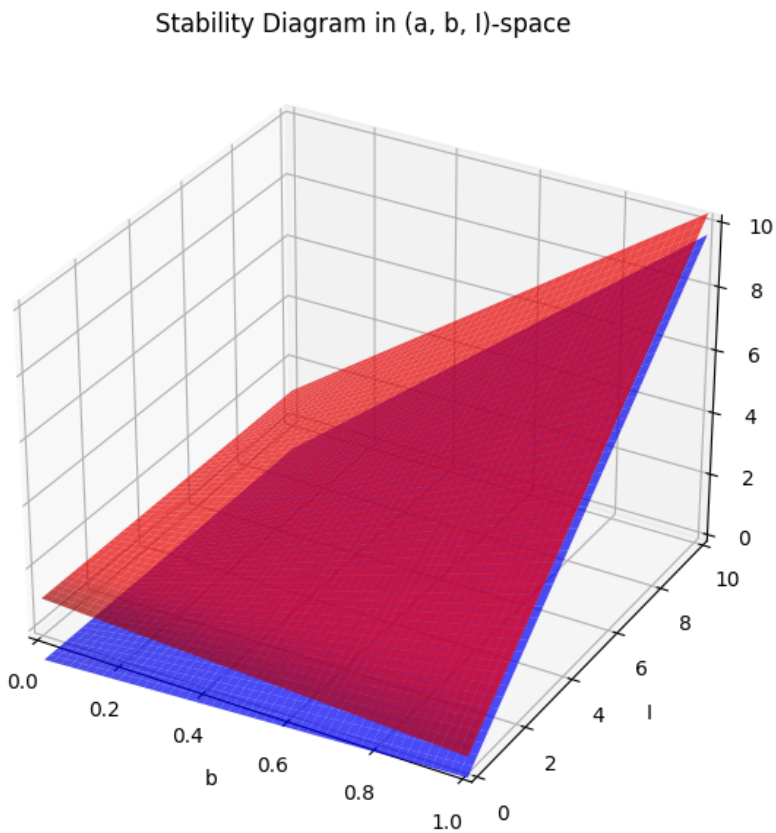
And setting $v_f = -1$ will determine the value for a^+ in terms of I and b, or the value for I^- in terms of a and b:

$$\begin{aligned} -1 + \frac{1}{3} + I &= \frac{-1+a}{b} \\ \implies a^+ &= b(I - \frac{2}{3}) + 1 \\ \implies I^- &= \frac{-1+a}{b} + \frac{2}{3} \end{aligned}$$

With these equations, we can construct a stability diagram, showing the regions of $(a, I)-$space that result in limit cycles. Below is one such diagram, for $b = 0.5$:



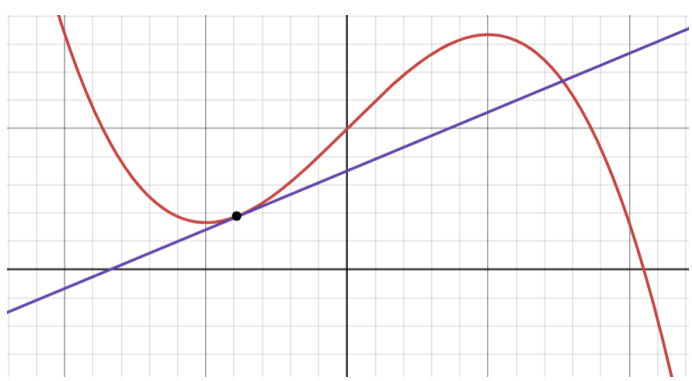
Since this region will also depend on b, a more elucidating visualization is to plot the equations in $(b, a, I)-$ space, where b is ranging from 0 to 1:



The volume between these two surfaces represents the parameter values of a, b, and I that will lead to a limit cycle and an unstable fixed point. Thus, what we see is actually two *codimension-3 supercritical Hopf bifurcations*. Note that as b approaches 1, the region shrinks, so limit cycles will become more rare.

5 $b > 1$: One or three fixed points

As we have seen, when $b > 1$, it is possible for the nullclines to intersect one or three points. Graphically, we see that changing the values of a, b and I continuously shift the nullclines until they meet. This intersection will necessarily first occur at a *point of tangency*, as shown below:



Thus, to find an equation governing the critical parameter values, we set the nullcline equations *and their derivatives* (with respect to v) equal to each other. The first equation is the same cubic equation we derived in section 3:

$$\frac{b}{3}v^3 - (b-1)v + a - bI = 0$$

The second equation is as follows:

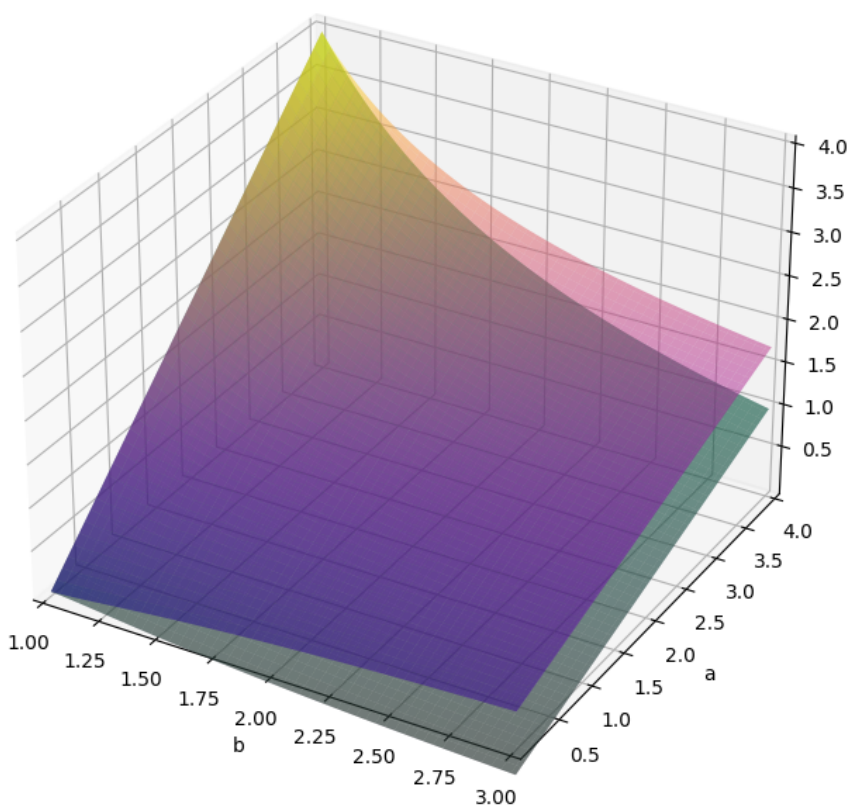
$$\frac{d}{dv}\left(v - \frac{v^3}{3} + I\right) = \frac{d}{dv}\left(\frac{v+a}{b}\right)$$
$$1 - v^2 = \frac{1}{b} \implies v = \pm\sqrt{\frac{b-1}{b}}$$

To derive a set of equations involving only a, b and I, we substitute these expressions for v into the first equation:

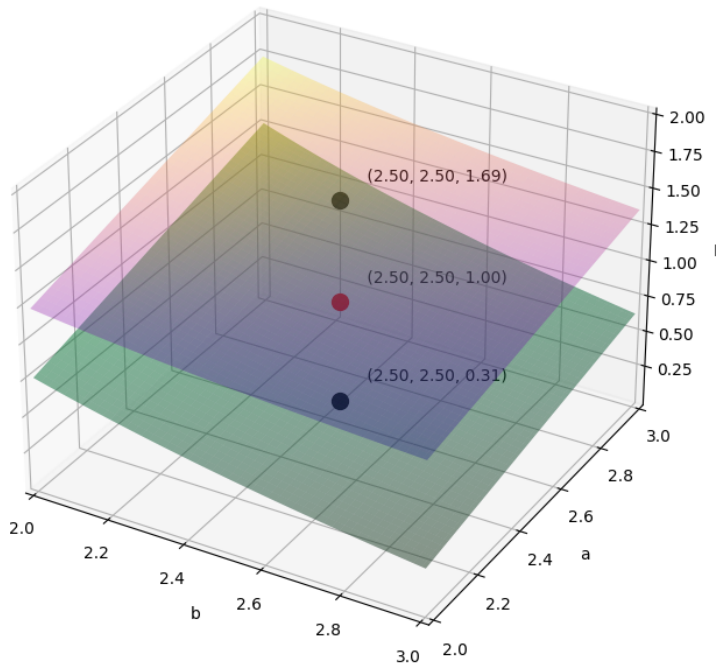
$$\frac{b}{3}\left(\pm\sqrt{\frac{b-1}{b}}\right)^3 - (b-1)\left(\pm\sqrt{\frac{b-1}{b}}\right) + a - bI = 0$$

5.1 Bifurcation and stability diagram

The derived equations are difficult to work with analytically, but we can visualize the region they map out in $(b, a, I)-$space:



Since each surface represents a point of tangency, the region in between the surfaces will consist of parameter values for which there exist 3 fixed points. Let's consider, for example, fixing b and a and varying I so that we enter and exist the region. Below are three such points plotted with the surface, where a and b have been fixed at 2.5, and values of I have been chosen below the region (0.31), inside the region (1) (shown in red) and above the region (1.69):



We can plot the trajectories of the system in the phase plane for each of these sets of parameter values to see what behaviors emerge.

- **Below region**: $a = 2.5$, $b = 2.5$, $I = 0.31$, $\epsilon = 0.01$

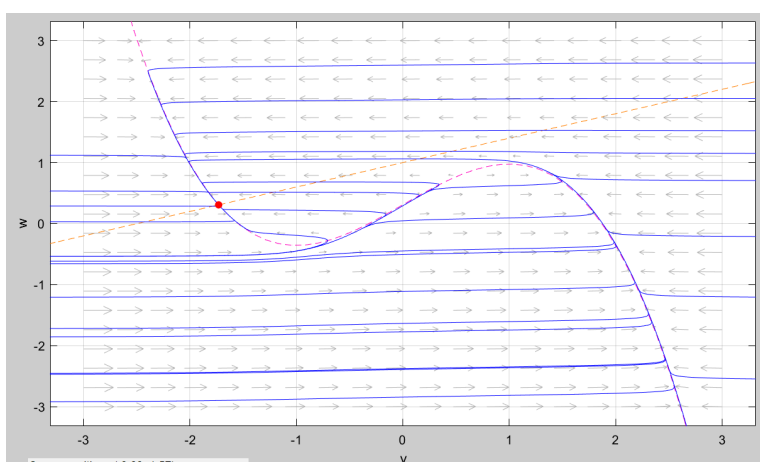


Figure 4: One stable fixed point on the left branch

- **Inside region**: $a = 2.5$, $b = 2.5$, $I = 1$, $\epsilon = 0.01$

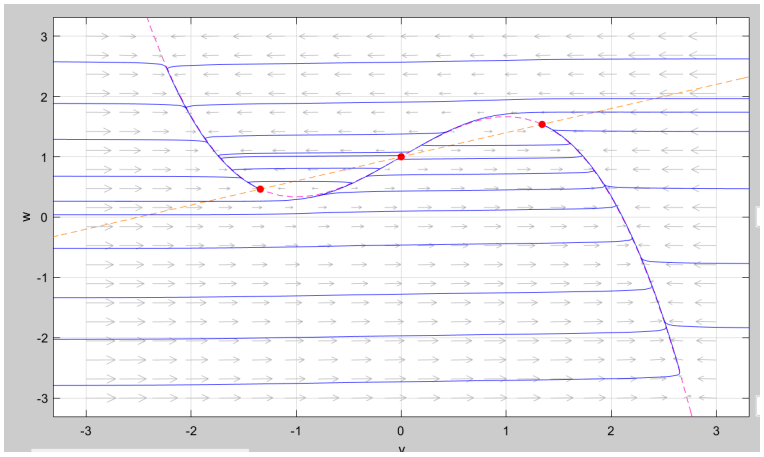


Figure 5: Two symmetrical fixed points and one unstable saddle point

- **Above region**: $a = 2.5$, $b = 2.5$, $I = 1.69$, $\epsilon = 0.01$

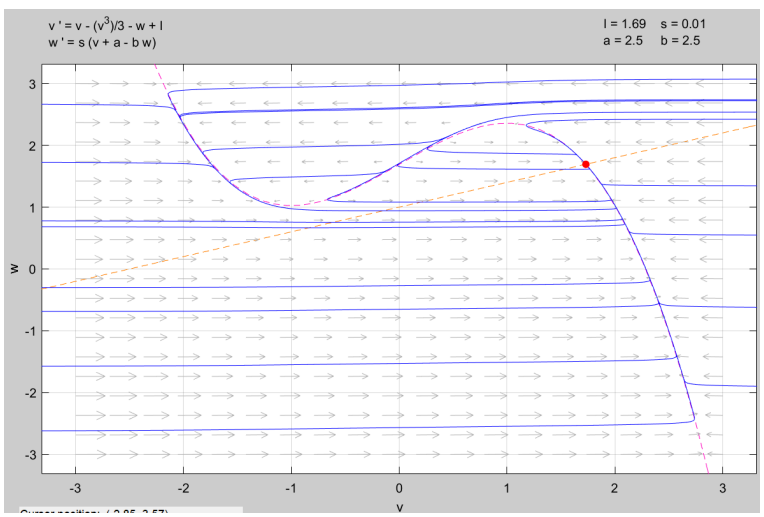


Figure 6: One stable fixed point on the right branch

Based on the behaviors in these plots, there are two *codimension-3 saddle-node bifurcations* for $b > 1$ which generate and then annihilate a pair of stable and unstable fixed points when the parameters are within a certain range. This range is bounded by the surfaces plotted previously.

6 Dynamical Regimes

Many sources categorize the various potential behaviors of this system in biological terms. Each region of parameter space corresponds to a certain *dynamical regime* of the neuron that is seen in practice (Nekorkin et al 2018). Understanding these dynamical regimes is essential for understanding the model's explanatory power.

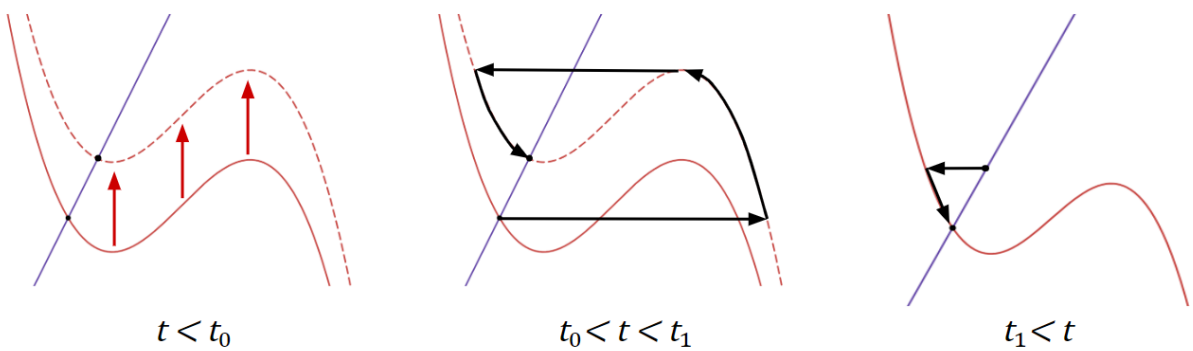
6.1 Excitability

Excitability is the ability of a neuron to fire excitation pulses (known as *action potentials*) in response to stimuli. A stimulus in this model corresponds to a pulse of current. For instance, suppose we let the parameter I be *time-dependent* by defining it as a rectangular function:

$$I(t) = \begin{cases} I_0 & \text{if } t \leq t_0, \\ I_s & \text{if } t \in [t_0, t_1], \\ I_0 & \text{if } t \geq t_1 \end{cases}$$

Where I_0 is a constant resting-state current, $(I_s - I_0)$ is the amplitude of the current pulse, and (t_0, t_1) represent the start and end times for the pulse. Now suppose $b < 0$ and $a > b(I_0 - \frac{2}{3}) + 1$. Then, as seen in section 4.1, there will initially be a stable fixed point (v_{f1}, w_{f2}) located on the left branch of the nullcline.

If we assume the neuron is initially in the resting state, then v and w will be at this fixed point. However, as soon as $t > t_0$, I will increase to I_s, shifting the cubic nullcline up and temporarily moving the fixed point to v_{f2}, w_{f2}, where $v_{f2} > v_{f1}$. Now the neuron is *below* the fixed point in phase space, and will quickly move towards the nullcline, eventually completing almost a full oscillation until settling at the new fixed point. After $t > t_1$, the neuron will return to its original resting state. This sequence of events is sketched below:



Thus, a pulse of current will cause the membrane voltage to spike and produce a single action potential. In this sense, the neuron is *excitable*.

6.2 Tonic Spiking

Tonic spiking refers to regularly spaced neuronal oscillations. This phenomenon is exhibited by the system when $b < 1$ and the fixed points lands in the middle branch. This will occur when the parameters are in the region between the Hopf bifurcation surfaces. The condition can be expressed as:

$$b(I - \frac{2}{3}) - 1 < a < b(I + \frac{2}{3}) + 1$$

Here there is no need for pulses of current; the oscillations will persist for constant current.

6.3 Excitation Block

Neurons often exhibit a behavior known as *excitation block*, where increasing the applied current actually suppresses neuronal output (somewhat counterintuitively) and prevents oscillations from taking place. This is the basis of medical treatments such as deep brain stimulation. Although the exact mechanism of action for excitation block is not known (Rosch & Markov 2004), the FitzHugh-Nagumo also exhibits this behavior as one of its dynamical regimes.

When $b < 1$ and $a < b(I + \frac{2}{3}) - 1$, there is a single stable fixed point located on the right branch of the nullcline. In this case, although the neuron is still excitable, there is no limit cycle so periodic oscillations are impossible. To see why this is an example of excitation block, consider the fact that increasing I above a certain threshold ($\frac{1+a}{b} - \frac{2}{3}$) will cause the fixed point to enter the right branch. Thus, increasing I moderately can create periodic oscillations, but increasing it *too much* will suppress them.

6.4 Bistability

Many neurons exhibit *bistability*: the coexistence of two modes of firing (Dovzhenok & Kuznetsov 2012). Usually, this involves two simultaneous kinds of spiking, such as tonic spiking and more irregular "bursting". The FitzHugh-Nagumo model also exhibits bistability, but between two stable resting states. This occurs when $b > 1$ and the parameters are within the region bounded by the saddle-node bifurcation surfaces:

$$\frac{b}{3}\left(\pm\sqrt{\frac{b-1}{b}}\right)^3 - (b-1)\left(\pm\sqrt{\frac{b-1}{b}}\right) + a - bI = 0$$

Although this dynamical regime has not been widely studied or linked to the observed behavior of neurons, the capacity for bistability demonstrates the flexibility the model.

7 Conclusion and Future Work

In this project, we analyzed the dynamical properties of the FitzHugh-Nagumo model, examining the existence and stability of fixed points and limit cycles and the bifurcations (Hopf and saddle-node) that occur in response to changes in the system parameters. We also demonstrated the existence of several dynamical regimes (excitability, tonic spiking, excitation block and bistability) and provided biological interpretations of the system's behavior.

The FitzHugh-Nagumo is a simple yet powerful model for the dynamics of a neuron. It also possesses enough flexibility to be applicable in a wide range of fields, from cardiology to algorithmic design. Despite its simplicity, it exhibits a rich and complex set of dynamical behaviors — indeed, too many to include in this paper.

In the future, a variety of further explorations are possible (and warranted), such as:

- Introducing stochastic noise in the form of Gaussian random variables to mimic the inherent noise of biological processes
- Considering discretely-coupled models to connect multiple instances of FHN neurons and examine synchronization phenomena (this was already studied in depth by Yanagita et al. (2005)).

8 Bibliography

1. Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of physiology*, 117(4), 500–544. https://doi.org/10.1113/jphysiol.1952.sp004764

2. J. Nagumo, S. Arimoto and S. Yoshizawa (1962). An Active Pulse Transmission Line Simulating Nerve Axon. *Proceedings of the IRE*, vol. 50, no. 10, pp. 2061-2070 10.1109/JRPROC.1962.288235.

3. FitzHugh R. (1961). Impulses and Physiological States in Theoretical Models of Nerve Membrane. *Biophysical journal*, 1(6), 445–466. https://doi.org/10.1016/s0006-3495(61)86902-6

4. Keener, J., & Sneyd, J. (2010). *Mathematical physiology: I: Cellular Physiology.* Springer Science & Business Media.

5. Kuramoto, Y., Battogtokh, D., & Nakao, H. (1998). Multiaffine chemical turbulence. Physical Review Letters, 81(16), 3543–3546. https://doi.org/10.1103/physrevlett.81.3543

6. Moayeri, M., Rad, J., & Parand, K. (2020). Dynamical behavior of reaction–diffusion neural networks and their synchronization arising in modeling epileptic seizure: A numerical simulation study. *Computers & Mathematics With Applications,* 80(8), 1887–1927. https://doi.org/10.1016/j.camwa.2020.08.020

7. Sherwood, W.E. (2014). FitzHugh–Nagumo Model. In: Jaeger, D., Jung, R. (eds) *Encyclopedia of Computational Neuroscience*. Springer, New York, NY. https://doi.org/10.1007/978-1-4614-7320-6_147-1

8. Nekorkin, V. I., Dmitrichev, A. S., asatkin, D. V., Klinshov, V. V., Kirillov, S. Y., Maslennikov, O. V., & Shchapin, D. S. (2018). Nonlinear dynamical models of neurons: Review. *Izvestiya VUZ Applied Nonlinear Dynamics*, 26(4), 5–58. https://doi.org/10.18500/0869-6632-2018-26-4-5-58

9. Grill, W. M., & Mclntyre, C. C. (2001). Extracellular excitation of central neurons: implications for the mechanisms of deep brain stimulation. *Thalamus & Related Systems*, 1(3), 269–277. doi:10.1017/S1472928801000255

10. Rosch, P. J., Markov, M. S. (2004). *Bioelectromagnetic Medicine*, 133-151. CRC Press

11. Dovzhenok, A., & Kuznetsov, A. S. (2012). Exploring neuronal bistability at the depolarization block. *PLoS ONE*, 7(8), e42811. https://doi.org/10.1371/journal.pone.0042811

12. Yanagita, T., Ichinomiya, T., & Oyama, Y. (2005). Pair of excitable FitzHugh-Nagumo elements: Synchronization, multistability, and chaos. *Physical Review E*, 72(5). https://doi.org/10.1103/physreve.72.056218